For this reason, the photon is sometimes described as the *messenger particle* of the electromagnetic force.

The gravitational field is also familiar since it constantly and consistently anchors us, and everything around us, to the earth's surface. As with electromagnetic fields, we are all immersed in a sea of gravitational fields; the earth's is dominant, but we also feel the gravitational fields of the sun, the moon, and the other planets. Just as photons are particles that constitute an electromagnetic field, physicists believe that *gravitons* are particles that constitute a gravitational field. Graviton particles have yet to be discovered experimentally, but that's not surprising. Gravity is by far the weakest of all forces (for example, an ordinary refrigerator magnet can pick up a paper clip, thereby overcoming the pull of the *entire* earth's gravity) and so it's understandable that experimenters have yet to detect the smallest constituents of the feeblest force. Even without experimental confirmation, though, most physicists believe that just as photons transmit the electromagnetic force (they are the electromagnetic force's messenger particles), gravitons transmit the gravitational force (they are the gravitational force's messenger particles). When you drop a glass, you can think of the event in terms of the earth's gravitational field pulling on the glass, or, using Einstein's more refined geometrical description, you can think of it in terms of the glass's sliding along an indentation in the spacetime fabric caused by the earth's presence, or—if gravitons do indeed exist—you can also think of it in terms of graviton particles firing back and forth between the earth and the glass, communicating a gravitational "message" that "tells" the glass to fall toward the earth.

Beyond these well-known force fields, there are two other forces of nature, the *strong nuclear force* and the *weak nuclear force,* and they also exert their influence via fields. The nuclear forces are less familiar than electromagnetism and gravity because they operate only on atomic and subatomic scales. Even so, their impact on daily life, through nuclear fusion that causes the sun to shine, nuclear fission at work in atomic reactors, and radioactive decay of elements like uranium and plutonium, is no less significant. The strong and weak nuclear force fields are called *Yang*-Mills fields after C. N. Yang and Robert Mills, who worked out their theoretical underpinnings in the 1950s. And just as electromagnetic